

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Justification

Annotation practices vary across datasets. Some are gold-standard manually annotated (DEFT-2021 [Q25], CLISTER [Q41], DiaMED [Q45] with documented IAA via Cohen's Kappa and Gwet's AC1 [Q46]), while others are silver-standard (CAS POS via Tagex 3 at 98% precision [Q42], ESSAI POS via TreeTagger [Q43]). Critically for the regulatory deployment, no dataset identifies a regulatory affairs specialist, pharmacovigilance officer, or EMA/ANSM legal expert as an annotator [Q45 — 'medical expert' is the closest, but is medical not regulatory]. Annotator demographics are clinical professionals, NLP researchers, and one medical expert — labels reflect clinical/research norms that may systematically diverge from regulatory-legal standards. Dataset analysis reveals concrete annotation-judgment artifacts: STS scores fail to flag legally consequential omissions (DEFT2020-D16, DEFT2020-D17), pharmaceutical company names mis-tagged as DISO (QUAERO-D18), drug names systematically untagged in E3C (E3C-D5, E3C-D13), and DOSAGE not distinguishing stated vs. prescribed vs. safety-threshold doses (DEFT2021-D14). With OC weighted HIGH, this represents a major convergent validity concern.

Strengths

- DEFT-2021 NER and MLC are gold-standard manually annotated [Q25]
- CLISTER STS scores averaged from multiple annotators [Q41]
- DiaMED has the most rigorously documented annotation process: 4 annotators, 2 sessions, IAA via Cohen's Kappa and Gwet's AC1 [Q45, Q46]
- Inter-annotator score arrays in DEFT2020 (DEFT2020-D7) provide visibility into annotator disagreement on safety-relevant content

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels do NOT systematically reflect regulatory affairs stakeholder perspectives. Annotators are clinical professionals, NLP researchers, or one medical expert [Q45] — none with EMA/ANSM regulatory affairs or pharmacovigilance expertise. The deployment's authoritative ground truth (regulatory specialists applying EMA QRD and ANSM circulars) is not represented in any dataset.

OC-2: Substantial disagreement between original annotators and regulatory population is empirically likely. DEFT2020 STS scoring fails to flag legally consequential omissions (DEFT2020-D16: omission of 'par mesure de précaution' scored as similar; DEFT2020-D17: loss of 'opiacé' qualifier scored 4.0). Variation type framework [WEB-6] indicates such omissions would trigger

formal Type IA/IB/II submissions — meaning regulatory annotators would score these substantially differently.

ID	Evidence
WEB-6	Variation Type IA/IB/II classification framework establishes legal significance of small text changes — regulatory annotators would score differently than clinical/NLP annotators
DEFT2020-D16	En conséquence, par mesure de précaution, il convient d'éviter d'allaiter..." vs. "Il convient d'éviter d'allaiter pendant la durée du traitement. — Omission of "par mesure de précaution" scored 2–5; regulatory significance not captured
DEFT2020-D17	Ce produit peut provoquer un syndrome de sevrage opiacé..." vs. "Ce produit peut entraîner un syndrome de sevrage s'il est administré moins de 4 heures après la prise d'un stupéfiant... — Loss of "opiacé" qualifier averaged as 4.0 — regulatory specificity difference not flagged

OC-3: Annotator demographics are partially documented: DiaMED reports 4 annotators including 1 medical expert [Q45]; CLISTER reports 'several annotators' [Q41]; CAS POS uses automatic Tagex 3 [Q42]; ESSAI uses automatic TreeTagger [Q43]. No Datasheets/Data Statements report regulatory affairs expertise. DiaMED's specific Kappa values were not retrievable from search results [WEB-2 reference].

ID	Evidence
WEB-2	DiaMED IAA Table 4 Kappa values not surfaced in search results — full paper PDF retrieval required

OC-4: Re-annotation by regulatory affairs annotator pool is necessary for regulatory deployment but not provided by the benchmark. The English-language ADE Eval task [WEB-12] achieved F1 ~0.79 with purpose-built tools and required modification — underscoring the regulatory annotation difficulty.

ID	Evidence
WEB-12	ADE Eval purpose-built English regulatory NLP achieved F1 ~0.79 — confirms difficulty of regulatory annotation alignment

OC-5: Aggregation methods include averaging STS scores into a single floating-point reference [Q41] — empirically, this can erase legally significant minority annotator concerns when most annotators score on general proximity rather than regulatory equivalence (DEFT2020-D7). Majority-class baselines for classification [Q56] do not handle minority regulatory-relevant categories.

ID	Evidence
DEFT2020-D7	scores: [5.0, 2.0, 4.0, 4.0, 5.0] — Wide annotator spread on safety instruction; relevant to human adjudication design

OC-6: Confirmed convergent validity and external validity concerns: (a) annotation authority gap is systemic across all 11 datasets analyzed; (b) silver-standard POS labels (CAS, ESSAI) propagate ~2% error rates at scale; (c) annotation gaps observed (E3C-D13 disease names tagged O); (d) at least one apparent annotation error (QUAERO-D18 company name as DISO); (e) DiaMED's Pan African medical expert applies sub-Saharan African clinical norms (DIAMED-D9, DIAMED-D12) rather than French regulatory norms.

ID	Evidence
QUAERO-D18	Richter Gedeon Eesti filiaal ... Polska Gedeon Richter Plc — Pharmaceutical company name annotated as DISO — likely annotation error
E3C-D13	Le patient a été mis sous traitement par ciclosporine avec une évolution rapide vers une leucémie aigue myéloblastique. — Clear disease name ("leucémie aigue myéloblastique") tagged O; suggests annotation gaps
DEFT2021-D14	l'infirmière anesthésiste se rend compte qu'elle a utilisé une ampoule de morphine de 10mg au lieu de 1 mg — Medication error case; DOSAGE annotation does not distinguish stated vs. prescribed vs. safety-threshold dose
DIAMED-D9	service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabé institution; confirms non-metropolitan-France provenance

Information Gaps

- Specific Kappa and Gwet's AC1 values from DiaMED Table 4 not surfaced in search [WEB-2]
- IAA for QUAERO, E3C, MORFITT, MANTRAGSC not reported

Requires Expert Verification

- Magnitude of regulatory-vs-clinical annotator disagreement on representative SmPC/PIL safety warning text
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Remediation

Gap 1: No annotator with regulatory affairs, pharmacovigilance, or EMA/ANSM submission expertise across any DrBenchmark dataset; ground truth reflects clinical/research norms

Recommendation: Recruit a regulatory affairs annotator pool (≥ 3 specialists with EMA QRD/ANSM submission experience) for re-annotation of a representative sample (e.g., 200 sentence pairs from DEFT2020 leaflet subset, 100 NER documents from QUAERO EMEA). Compute adjusted IAA against original annotations to quantify the regulatory-vs-clinical divergence empirically.